DESIGN AND SIMULATION OF AN AUTOMATIC STREET LIGHT CONTROL SYSTEM

Subject: Digital Design and Computer Architecture

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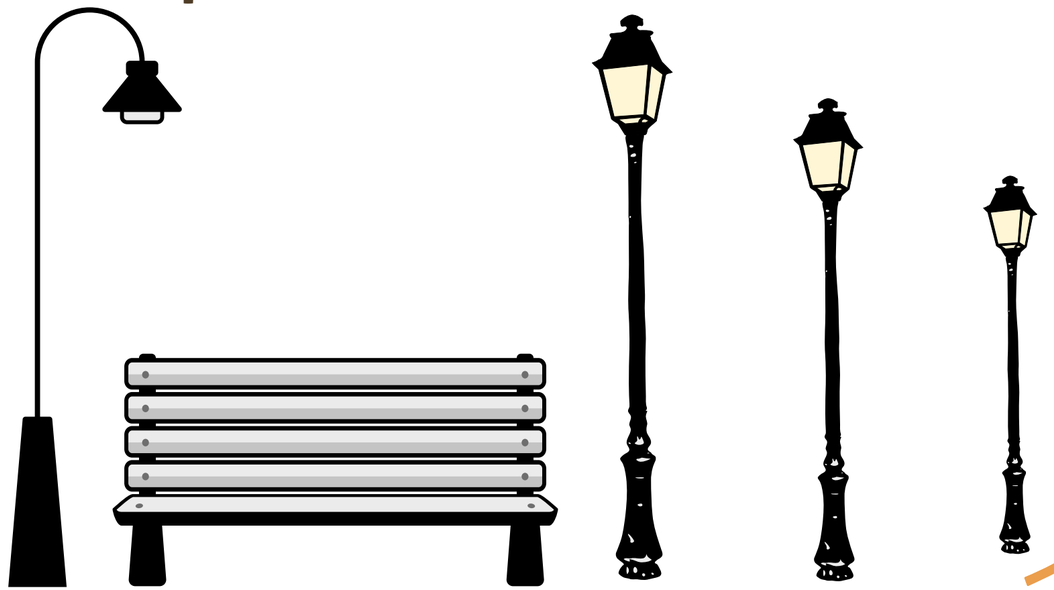
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SECTION – 2 GROUP 9



Introduction

- Automatic street lights operate according to the surrounding light conditions.
- They turn ON during darkness and OFF during daylight.
- The system reduces unnecessary electricity consumption and manual effort.
- Digital logic is used to represent the control operation.

Objectives

- To design an automatic street light control circuit.
- To implement the circuit using logic gates.
- To simulate the circuit using Logisim Evolution.
- To verify the output using a truth table.

Purpose.

In this project, the light sensor condition is represented using a digital input in Logisim.

Basic Concept

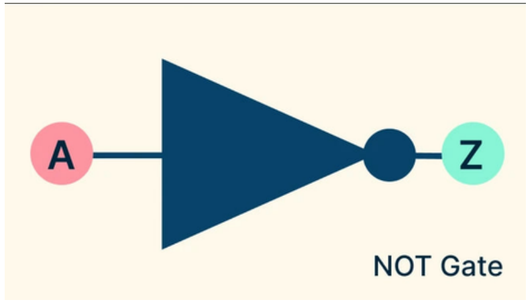
- Input D represents the daylight condition.
- D = 1 indicates daylight is detected.
- D = 0 indicates darkness is detected.
- Output L controls the street light indicator.
- A NOT gate reverses the input signal.

Boolean Expression

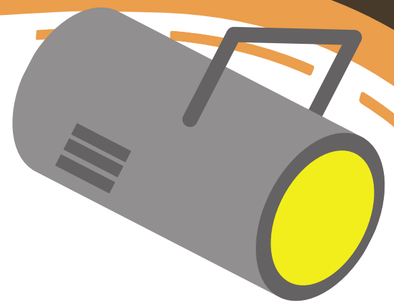
$L = \neg D$

Where:

- **A = 1 → Street light ON**
- **A = 0 → Street light OFF**



Circuit Implementation



The circuit is designed and tested using Logisim, a digital logic simulation tool.

Components Used

1. Input pin to represent the sensor.
2. Output pin to represent the street light.
3. Wire to connect the input and output.
4. Optional buffer gate to represent the signal path

Working of the Circuit

1. The input pin represents the light sensor's digital signal.
2. When the input is set to 0, the output becomes 0 and the street light remains OFF.
3. When the input is set to 1, the output becomes 1 and the street light turns ON.
4. The simulation verifies the relationship between the input and output.

Truth Table

Input D Condition

Output L Street Light

Daylight Input (D)	Darkness (~D)	Lamp Output (L)
0	1	1 - ON
1	0	0 - OFF

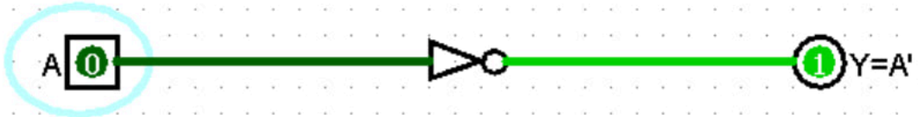
Simulation Observations

1. When Input D = 0, Output L = 1.
2. When Input D = 1, Output L = 0.
3. The output follows the input, verifying the Boolean expression

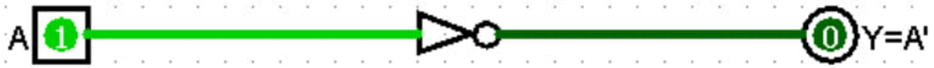
   Y = ~A

Logic Circuit :

- When input is 0.



- When input is 1.



- Truth Table:

A	YA
0	1
1	0

Conclusion

- Successfully designed a basic automatic street light control logic circuit.
- Verified its operation using a NOT gate and truth table.
- Demonstrated automatic switching through Logisim Evolution.

Applications

- Street lighting systems
- Parking areas
- Garden and outdoor lights
- Building corridors
- Security lighting

Future Improvements

- Use an LDR sensor to detect actual light intensity.
- Add a motion sensor to control lighting based on movement.
- Include a timer and adjustable brightness threshold.
- Use a transistor or relay driver for practical lamp control.


THANK YOU